Lecture 1: Difference Equation

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1. Difference operators

$$\text{1-st order difference:} \qquad \Delta y_t = y_t - y_{t-1}$$

$$\text{2-nd order difference:} \qquad \Delta^2 y_t = \Delta(\Delta y_t) = \Delta(y_t - y_{t-1}) = \Delta y_t - \Delta y_{t-1} = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2})$$

$$= y_t - 2y_{t-1} + y_{t-2}.$$

2. Period

Period: an interval of time $\Rightarrow$ y_t is the y for t^{th} interval of time, not for a point of time.

* t 是一个时间段概念，不是时间点概念

$$\Delta y_t = y_t - y_{t-1}$$

is the change from $(t-1)^{\text{th}}$ time interval to t^{th} interval, not a derivative at a point of time t.

* Δy_t 是变化量，并不是在 t 点的微分（变化率）

$$\text{Derivative} = \dot{y}_t = \frac{\Delta y_t}{y_{t-1}} = \frac{y_t - y_{t-1}}{y_{t-1}} = \frac{y_t}{y_{t-1}} - 1 \approx \log\left(\frac{y_t}{y_{t-1}}\right)$$

$$\text{when } \frac{y_t - y_{t-1}}{y_{t-1}} \text{ is small, } \log\left(1 + \frac{y_t}{y_{t-1}} - 1\right) \approx \frac{y_t}{y_{t-1}} - 1 = \frac{y_t - y_{t-1}}{y_{t-1}}.$$

3. Solve a first-order difference equation

Solve a first-order difference equation $\Longleftrightarrow$ rewrite y_t as a function of t.

Example:

$$y_t + a y_{t-1} = C.$$

(1) Brutal force: iterative method

$$y_t = -ay_{t-1} + C$$
$$y_{t-1} = -ay_{t-2} + C$$

Hence

$$y_t = a^2 y_{t-2} - aC + C, \qquad y_{t-2} = -ay_{t-3} + C.$$

Continuing,

$$y_t = -a^3 y_{t-3} + a^2 C - aC + C - \cdots$$

and in general,

$$y_t = (-a)^t y_0 + C \sum_{j=0}^{t-1} (-a)^j.$$

If $a \neq -1$,

$$\sum_{j=0}^{t-1} (-a)^j = \frac{1 - (-a)^t}{1 + a}.$$

So

$$y_t = (-a)^t \left(y_0 - \frac{C}{1+a} \right) + \frac{C}{1+a}, \qquad a \neq -1.$$

Equivalently,

$$y_t = A(-a)^t + \frac{C}{1+a}, \qquad a \neq -1.$$

If $a = -1$,

$$\sum_{j=0}^{t-1} (-a)^j = \sum_{j=0}^{t-1} 1 = t \qquad \Rightarrow \qquad y_t = y_0 + Ct.$$

Thus

$$y_t = \begin{cases} A(-a)^t + \dfrac{C}{1+a}, & a \neq -1, \\ Ct + y_0, & a = -1, \end{cases} \qquad \text{where } A = y_0 - \frac{C}{1+a}.$$

(2) General method

Since the system is *linear*, we can separate:

1) the part driven by constant C,

2) the part comes from the dynamics.

Why $y_t = y_t^p + y_t^c$ is valid?

Define linear operator

$$L(\cdot) = (\cdot)_t + a(\cdot)_{t-1}.$$

Then

$$y_t + ay_{t-1} = C \qquad \Rightarrow \qquad L(y) = C.$$

If y^p is any particular solution such that

$$L(y^p) = C,$$

and if y^c is any solution of the homogeneous equation

$$L(y^c) = 0 \qquad \text{(complementary solution)},$$

then by linearity,

$$L(y^p + y^c) = L(y^p) + L(y^c) = C + 0 = C.$$

Therefore,

$$y^p + y^c$$

solves the equation

$$y_t + ay_{t-1} = C.$$

$$\boxed{\text{All solutions} = \text{one particular solution} + \text{all homogeneous solutions.}}$$

Now we solve the difference equation with general method.

1) Complementary solution Since exponential sequences are eigenfunctions of the lag operator, try

$$y_t = Ab^t.$$

Then for the homogeneous equation,

$$Ab^t + aAb^{t-1} = 0 \quad \Rightarrow \quad b + a = 0 \quad \Rightarrow \quad b = -a.$$

Hence

$$y_t^c = A(-a)^t.$$

2) Particular solution Since the forcing term is a constant C, we guess

$$y_t^p = k.$$

Then

$$k + ak = C \quad \Rightarrow \quad k = \frac{C}{1+a}, \qquad a \neq -1.$$

If $a = -1$,

$$y_t - y_{t-1} = C \quad \Rightarrow \quad y_t^p = Ct.$$

Therefore,

$$y_t = y_t^c + y_t^p = \begin{cases} A(-a)^t + \dfrac{C}{1+a}, & a \neq -1, \\ A + Ct, & a = -1. \end{cases}$$

Using the initial condition:

If $a \neq -1$,

$$y_0 = A(-a)^0 + \frac{C}{1+a} = A + \frac{C}{1+a} \quad \Rightarrow \quad A = y_0 - \frac{C}{1+a}.$$

If $a = -1$,

$$y_0 = A \quad \Rightarrow \quad A = y_0.$$

4. Dynamic stability of equilibrium

Consider

$$y_t + ay_{t-1} = C.$$

The steady state is defined by

$$y_t = y_{t-1} = \bar{y},$$

so

$$\bar{y} + a\bar{y} = C \quad \Rightarrow \quad \bar{y} = \frac{C}{1+a} \qquad (a \neq -1).$$

Subtract steady state from the original equation:

$$y_t - \bar{y} + a(y_{t-1} - \bar{y}) = C - \bar{y} - a\bar{y}.$$

Define deviation from steady state as

$$\tilde{y}_t = y_t - \bar{y}.$$

Then

$$\tilde{y}_t + a\tilde{y}_{t-1} = 0 \quad \Rightarrow \quad \tilde{y}_t = (-a)\tilde{y}_{t-1}.$$

Let

$$b = -a.$$

Then

$$\tilde{y}_t = b\tilde{y}_{t-1} \quad \Rightarrow \quad \tilde{y}_t = b^t \tilde{y}_0.$$

* $b = -a$ controls convergence 收斂性

$$\begin{cases} |b| < 1 \Rightarrow b^t \to 0, \ \tilde{y}_t \to 0, \ y_t \to \bar{y} & \text{(convergence)}, \\ |b| > 1 \Rightarrow b^t \to \infty, \ \tilde{y}_t \to \infty & \text{(divergence)}. \end{cases}$$

5. Sign of b and oscillation / monotonicity

Sign of b also controls oscillation 交替振荡性 / monotonicity 单调性.

$$\begin{cases} b > 0: & b^t \text{ keeps the same sign for all } t \Rightarrow \tilde{y}_t \text{ keeps the same sign for all } t, \\ & \text{path is non-oscillatory (monotone, towards / away from } \bar{y}). \\ b < 0: & b^t \text{ alternates sign } \Rightarrow \tilde{y}_t \text{ flips sign each period} \\ & \Rightarrow \text{oscillate around } \bar{y}. \end{cases}$$

Further:

$$\begin{cases} b > 1, & y_t \text{ monotonically diverges from } \bar{y}, \\ b = 1, & y_t \text{ keeps the same distance away from } \bar{y}, \\ 0 < b < 1, & y_t \text{ monotonically converge to } \bar{y}, \\ b = 0, & y_t \text{ stays at steady state } \bar{y} \text{ always}, \\ -1 < b < 0, & y_t \text{ oscillate around and converge to } \bar{y}, \\ b = -1, & y_t \text{ oscillate around } \bar{y} \text{ and keeps same absolute distance}, \\ b < -1, & y_t \text{ oscillate around and diverge from } \bar{y}. \end{cases}$$

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6. The role of A

Since the general solution can be written as

$$y_t = \bar{y} + Ab^t \quad \Rightarrow \quad A = y_0 - \bar{y}$$

(initial deviation).

Scale effect:

$$|A| \uparrow \Rightarrow \text{start farther from equilibrium} \Rightarrow \text{bigger movements along path.}$$

Mirror effect:

$$A > 0: \text{ start above } \bar{y}; \qquad A < 0: \text{ start below } \bar{y}.$$

7. Example: Inventory model

Example: Inventory model.

$$Q_{d,t} = \alpha - \beta p_t \qquad \text{demand}$$

$$Q_{s,t} = -\gamma + \delta p_t \qquad \text{supply}$$

$$p_{t+1} = p_t - \sigma\left(Q_{s,t} - Q_{d,t}\right)$$

$$\alpha, \beta, \gamma, \delta, \sigma > 0 \qquad \text{solve for } p_t\text{'s path}$$

Then

$$Q_{s,t} - Q_{d,t} = -(\alpha + \gamma) + (\beta + \delta) p_t.$$

Plug in:

$$p_{t+1} = \big[1 - \sigma(\beta + \delta)\big] p_t + \sigma(\alpha + \gamma).$$

At steady state,

$$p_{t+1} = p_t = \bar{p} \quad \Rightarrow \quad \bar{p} = \frac{\alpha + \gamma}{\beta + \delta}.$$

Thus, the deviation dynamics is

$$p_t - \bar{p} = \big[1 - \sigma(\beta + \delta)\big](p_{t-1} - \bar{p}).$$

Stability condition tells us:

$$|1 - \sigma(\beta + \delta)| < 1 \quad \Rightarrow \quad 0 < \sigma(\beta + \delta) < 2,$$

system converges.

Oscillation condition tells us:

$$1 - \sigma(\beta + \delta) < 0 \quad \Rightarrow \quad \sigma(\beta + \delta) > 1$$

creates oscillation.

* Economic intuition: if price adjustment is too aggressive $\sigma \uparrow\uparrow$, creates oscillation and divergence.

If supply or demand elasticity in terms of price too high

$$\beta \uparrow\uparrow / \delta \uparrow\uparrow$$

$\Rightarrow$ creates oscillation or even divergence.

8. Higher-order difference equations

2-nd order linear difference equation with constant coefficients & constant term:

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = C.$$

1° Particular solution y^p: the intertemporal equilibrium level

Try a constant:

$$y^p = k \quad \Rightarrow \quad k(1+a_1+a_2) = C \quad \Rightarrow \quad k = \frac{C}{1+a_1+a_2}, \qquad 1+a_1+a_2 \neq 0.$$

Try a first-degree polynomial:

$$y^p = kt.$$

Then

$$k(t+2) + a_1 k(t+1) + a_2 kt = C$$

$$\Rightarrow \quad k\big[(1+a_1+a_2)t + (2+a_1)\big] = C.$$

Impose

$$1+a_1+a_2 = 0$$

then

$$k = \frac{C}{2+a_1}, \qquad 2+a_1 \neq 0.$$

Try a second-degree polynomial:

$$y^p = kt^2.$$

Then

$$k(t+2)^2 + a_1 k(t+1)^2 + a_2 kt^2 = C.$$

The notes impose

$$\begin{cases} 1+a_1+a_2 = 0, \\ 2+a_1 = 0, \end{cases} \qquad \Rightarrow \qquad \begin{cases} a_1 = -2, \\ a_2 = 1. \end{cases}$$

Substituting:

$$k(t^2+4t+4) - 2k(t^2+2t+1) + kt^2 = C$$

$$\Rightarrow \quad 2k = C \quad \Rightarrow \quad k = \frac{C}{2}.$$

Therefore

$$y^p = \begin{cases} \dfrac{C}{1+a_1+a_2}, & 1+a_1+a_2 \neq 0, \\ \dfrac{C}{2+a_1}\,t, & 1+a_1+a_2 = 0,\ a_1 \neq -2, \\ \dfrac{C}{2}\,t^2, & a_1 = -2,\ a_2 = 1. \end{cases}$$

2° Complementary solution y^c: deviation from the equilibrium in each period

Look at the homogeneous equation:

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = 0.$$

Try

$$y_t = Ab^t.$$

Then

$$b^2 + a_1 b + a_2 = 0.$$

Case 1: Two distinct real roots If

$$a_1^2 - 4a_2 > 0,$$

then

$$y^c = A_1 b_1^t + A_2 b_2^t$$

where

$$b_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}, \qquad b_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}.$$

Case 2: Repeated real root If

$$a_1^2 - 4a_2 = 0,$$

then

$$b = b_1 = b_2 = -\frac{a_1}{2},$$

and

$$y^c = A_3 b^t + A_4 t b^t.$$

Case 3: Complex roots If

$$a_1^2 - 4a_2 < 0,$$

then

$$b_{1,2} = h \pm vi, \qquad h = -\frac{a_1}{2}, \qquad v = \frac{\sqrt{4a_2 - a_1^2}}{2}.$$

So

$$y^c = A_5 (h + vi)^t + A_6 (h - vi)^t.$$

Using De Moivre's Theorem:

$$(h \pm vi)^t = R^t \big(\cos(\theta t) \pm i \sin(\theta t) \big)$$

where

$$R = \sqrt{h^2 + v^2} = \sqrt{a_2}, \qquad \cos\theta = \frac{h}{R} = -\frac{a_1}{2\sqrt{a_2}}, \qquad \sin\theta = \frac{v}{R} = \sqrt{1 - \frac{a_1^2}{4a_2}}.$$

Hence

$$y^c = R^t\big(A_5 \cos\theta t + A_6 \sin\theta t\big),$$

where

$$R = \sqrt{a_2}, \qquad \theta \text{ is defined as } \begin{cases} \cos\theta = -\dfrac{a_1}{2\sqrt{a_2}}, \\ \sin\theta = \sqrt{1 - \dfrac{a_1^2}{4a_2}}. \end{cases}$$

9. Convergence under higher-order difference equation

1° Distinct root: b_{dom} is the root with highest absolute value. Convergence requires

$$|b_{\text{dom}}| < 1.$$

2° Repeated root:

$$|b| < 1 \Rightarrow \text{converges}.$$

3° Complex root:

$$R > 1: \text{ explosive oscillations}, \qquad R < 1: \text{ damped oscillations}.$$

10. Simultaneous difference equation (state-space form)

Let

$$x_t = y_{t+1}.$$

Then

$$\begin{cases} x_{t+1} + a_1 x_t + a_2 y_t = C, \\ x_t = y_{t+1}. \end{cases}$$

Therefore,

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} C \\ 0 \end{pmatrix}.$$

Generalize:

$$X_{t+1} = AX_t + B$$

where

$$X_t = (x_{1t}, x_{2t}, \dots, x_{nt})$$

is state space.

$$A \text{ is a constant matrix with } a_{ij} \text{ as elements } (i, j = 1, \dots, n),$$

$$B \text{ is an array with } B_i,\ i = 1, \dots, n \text{ as elements.}$$

In this example,

$$X_{t+1} = \begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix}, \qquad A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} C \\ 0 \end{pmatrix}.$$

Everything about dynamics and stability is controlled by eigenvalues of A.
Ignore constant term B to study deviation from steady state:

$$X_{t+1} = AX_t.$$

Try solution:

$$X_t = vb^t.$$

Plug in:

$$vb^{t+1} = A(vb^t) \quad \Rightarrow \quad bv = Av.$$

Here b is an eigenvalue of A, with eigenvector v.
A nonzero v exists iff

$$\det(A - bI) = 0.$$

11. Characteristic equation

Characteristic equation:

$$p(b) = |A - bI| = 0.$$

For a 2×2 matrix:

$$p(b) = b^2 - Tb + D = 0,$$

where

$$T = \text{trace}(A), \qquad D = \det(A).$$

Also,

$$T = b_1 + b_2, \qquad D = b_1 b_2,$$

so

$$p(b) = b^2 - (b_1 + b_2)b + b_1 b_2 = (b - b_1)(b - b_2).$$

Eigenvalues are the solution of the characteristic equation: b_1, b_2.
Back to the example, since

$$A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix},$$

we have

$$T = \mathrm{tr}(A) = -a_1, \qquad D = \det(A) = a_2.$$

Thus characteristic equation:

$$b^2 + a_1 b + a_2 = 0.$$

Eigenvalues:

$$\frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}, \qquad \text{if } a_1^2 > 4a_2.$$

12. Stability triangle on the (T, D) plane

1°

$$\Delta = T^2 - 4D.$$

$$\begin{cases} \Delta < 0: & \text{complex roots } \Rightarrow \text{above red curve,} \\ \Delta > 0: & \text{real roots } \Rightarrow \text{below red curve.} \end{cases}$$

2°

$$p(1) = 0, \quad p(-1) = 0$$

are the black curves.

$$p(1) = 1 - T + D, \qquad \text{right to } p(1) = 0 \Rightarrow p(1) < 0.$$

$$p(-1) = 1 + T + D, \qquad \text{left to } p(-1) = 0 \Rightarrow p(-1) < 0.$$

3°

$$D = 1 \text{ and } |b| = 1 \text{ for complex roots}$$

is the green curve.

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Why D controls $|b|$ in complex case?

If roots are complex conjugates

$$b_{1,2} = re^{\pm i\theta} \quad \Rightarrow \quad D = b_1 b_2 = r^2 \quad \Rightarrow \quad r = \sqrt{D}.$$

Stable $(r < 1)$ implies

$$D < 1.$$

Thus, above green curve $|b| > 1$, below green curve $|b| < 1$.

* Conditions for two stable roots (both inside unit circle)

$$D < 1 \quad \text{and} \quad p(1) > 0 \quad \text{and} \quad p(-1) > 0$$
$$\Rightarrow \quad D < 1, \quad 1 - T + D > 0, \quad 1 + T + D > 0$$
$$\Rightarrow \quad D < 1 \quad \text{and} \quad |T| < 1 + D.$$

$$\begin{cases} \text{complex roots with modulus（模）} < 1, & \text{if } \Delta < 0 \ (T^2 < 4D), \\ \text{real roots both between } -1 \text{ and } 1, & \text{if } \Delta > 0 \ (T^2 > 4D). \end{cases}$$

* Conditions for one stable root and one unstable root.

$$p(1) < 0 \ \& \ p(-1) > 0 \qquad \text{or} \qquad p(1) > 0 \ \& \ p(-1) < 0.$$

That is,

$$\begin{cases} 1 - T + D < 0, \\ 1 + T + D > 0, \end{cases} \qquad \text{or} \qquad \begin{cases} 1 - T + D > 0, \\ 1 + T + D < 0. \end{cases}$$

Thus,

$$|T| > |1 + D|.$$

$$p(1) \ \& \ p(-1) \text{ has opposite signs } \Rightarrow \text{There is a root in the interval } (-1, 1).$$

Since

$$p(1) < 0, \ p(b) \to +\infty \text{ as } b \to +\infty$$

or

$$p(-1) < 0, \ p(b) \to +\infty \text{ as } b \to -\infty,$$

there is another root outside the interval $[-1, 1]$.

* Conditions for explosive and unstable roots (both roots outside unit circle or complex with modulus > 1)

Four cases:

1° $p(1) < 0, \ p(-1) < 0, \ D < 0, \ |T| > 1$

2° $p(1) > 0, \ p(-1) > 0, \ D > 1, \ T < -2$

3° $\Delta < 0$ and $D > 1$

4° $p(1) > 0, \ p(-1) > 0, \ D > 1, \ T > 2$

13. Region-by-region summary

Region	Conditions	Roots
1	$p(1) < 0$ & $p(-1) > 0$	saddle: $\begin{cases} 1 \text{ stable root in } (-1,1), \\ 1 \text{ unstable root in } (1,+\infty) \end{cases}$
2	$p(1) < 0$ & $p(-1) < 0$	explosive: $\begin{cases} 1 \text{ unstable root in } (1,+\infty), \\ 1 \text{ unstable root in } (-\infty,-1) \end{cases}$
3	$p(1) > 0$ & $p(-1) < 0$	saddle: $\begin{cases} 1 \text{ stable root in } (-1,1), \\ 1 \text{ unstable root in } (-\infty,-1) \end{cases}$
4	$p(1) > 0$ & $p(-1) > 0$ & $\Delta > 0$ & $D > 1$ $(T < -2)$	explosive: 2 real unstable roots in $(-\infty,-1)$
5	$p(1) > 0$ & $p(-1) > 0$ & $\Delta < 0$ & $D > 1$	explosive: complex conjugates with modulus $r = \sqrt{D} > 1$, oscillates & diverges
6	$p(1) > 0$ & $p(-1) > 0$ & $\Delta < 0$ & $D < 1$ $(-2 < T < 2)$	stable: complex conjugates with modulus $r = \sqrt{D} < 1$, oscillates & converges
7	$\Delta > 0$ & $D < 1$ & $p(1) > 0$ & $p(-1) > 0$	stable: 2 real roots $\lvert b_1 \rvert < 1$, $\lvert b_2 \rvert < 1$
$7a$	$D > 0$ & $T < 0$	$-1 < b_2 \leq b_1 < 0$ (same sign, and $T = b_1 + b_2$ negative) decay but can flip sign
$7b$	$D < 0$	$-1 < b_2 < 0 < b_1 < 1$ (opposite sign, $D = b_1 b_2$)decay but can flip sign
$7c$	$D > 0$ & $T > 0$	$0 < b_2 \leq b_1 < 1$ (same sign, and $T = b_1 + b_2$ positive) monotone converge
8	$\Delta > 0$, $D > 1$, $T > 2$ $p(1) > 0$, $p(-1) > 0$	explosive: 2 real roots $b_1 > 1$, $b_2 > 1$

14. Blanchard and Kahn (BK) conditions

Predetermined variable: initial value is given, states like k, h, b (i.e. capital, hour, bond).
Jump variable: initial value not given, can jump to satisfy equilibrium, like c, p.

$$\text{Stable roots} = \#\text{ predetermined variables} \Rightarrow \text{Saddle path (Determinacy)}$$

$$\text{Stable roots} < \#\text{ predetermined variables} \Rightarrow \text{Source (Explosive, no convergent equilibrium)}$$

Stable roots > # predetermined variables ⇒ Sink (Indeterminacy)

15. Why?

Linearize around steady state, we write deviations as a first-order system:

$$X_{t+1} = AX_t.$$

If A has eigenpairs (b_i, v_i), the general solution can be written as sum of modes:

$$X_t = \sum_i \alpha_i b_i^t v_i.$$

If

$$|b_i| < 1,$$

mode dies out ⇒ stable.

If

$$|b_i| > 1,$$

mode blows up ⇒ unstable / explosive.

$$\text{A convergent equilibrium path must impose: } \alpha_i = 0 \quad \text{for every } |b_i| > 1,$$

which is a set of restrictions, one restriction per unstable root.

Now partition variables into

$$\{\text{predetermined states } s_t \text{ (given } s_0) \text{ with dimension } n_s,$$

$$\text{jump variables } j_t \text{ (free } j_0) \text{ with dimension } n_j\}.$$

$$X_t = (s_t, j_t).$$

Initial conditions pin down s_0, but can choose j_0.

Let

$$n_u = \#\text{ unstable roots}, \qquad n_\ell = \#\text{ stable roots}.$$

$$n_u + n_\ell = n_s + n_j.$$

To satisfy n_u restrictions, we have n_j free choices to satisfy them. Thus

$$n_u = n_j \qquad \Longleftrightarrow \qquad n_\ell = n_s.$$

Case A

$$n_\ell = n_s \quad (n_u = n_j):$$

jump variables jump to land on saddle path.

Case B

$$n_\ell < n_s \quad (n_u > n_j):$$

too many unstable modes to eliminate, but too few jump variables to adjust.

Case C

$$n_\ell > n_s \quad (n_u < n_j):$$

more jump freedom needed to kill unstable modes (multiple equilibria)